

Relay

How the relay works

Relay carries your message to a station you cannot reach directly, by passing it through a third station that can reach you both. That middle station is called the relay station.

JS8Map finds the best relay station for you, based on propagation as it is at that moment. It ranks the candidates on mutual SNR reports. Then it draws the message pathway traveling, hop by hop, so you can see how far it has got.

Starting a relay

Enter the target and choose an audience

The relay controls sit at the bottom right of the JS8Map control window. Type the callsign you want to reach in the Relay box. In the box beside it choose who to ask — AllCall, which asks everyone hearing you, or one of your JS8Call groups, which asks only that group. Then click Search.

JS8Map builds the correct command in JS8Call for you. Everything that comes back is sorted and placed across the top of the HTML map.

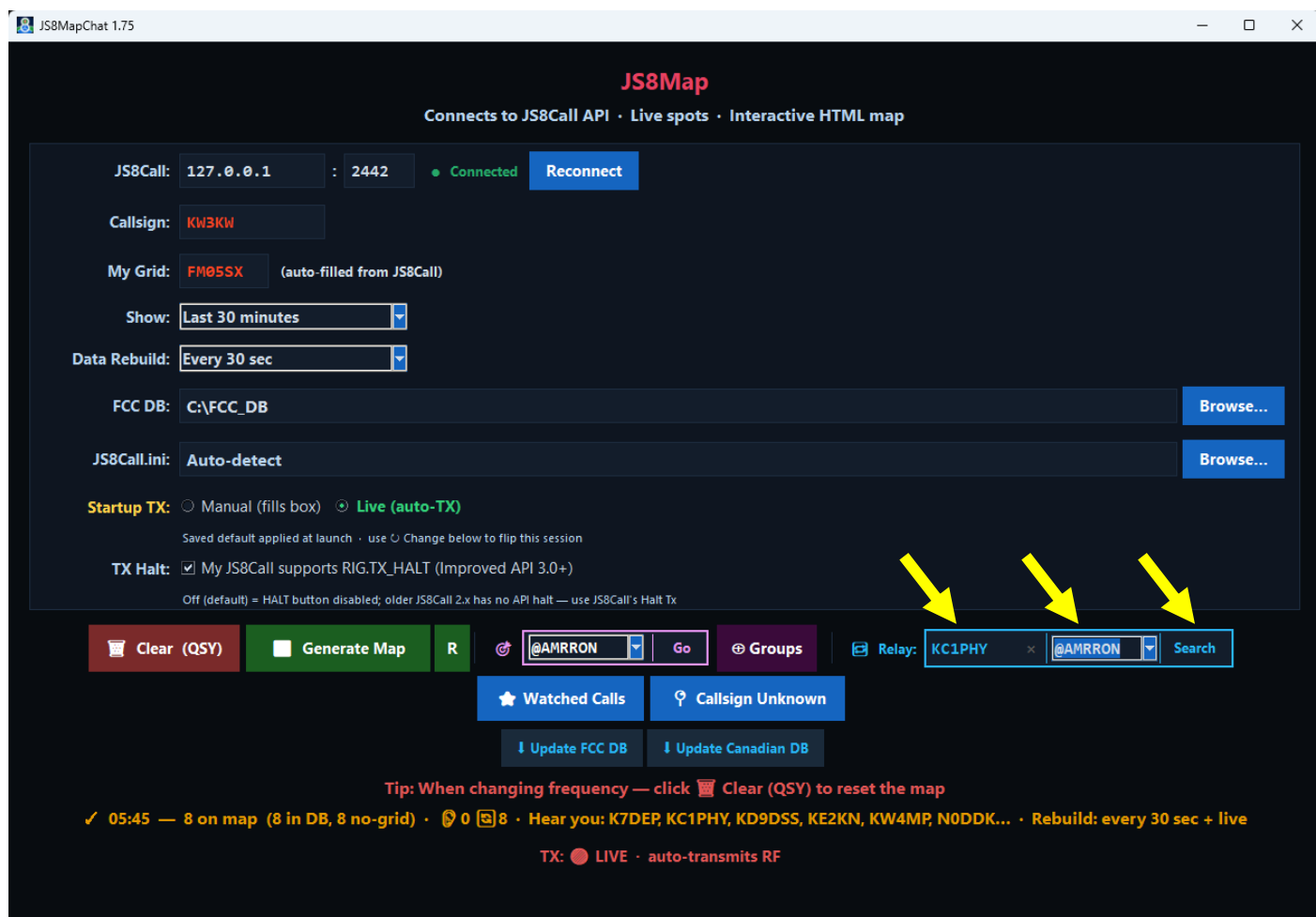


Figure 1 — The relay controls in the JS8Map control window. Target callsign, audience, Search. — updated image

PREFER A GROUP OVER ALLCALL

Not every operator allows their rig to be used for relay, and there is no way to tell from the outside which ones do. AllCall will therefore turn up stations that answer but never forward. Most groups **run the relay feature** and have it switched on, so choosing a group gives a noticeably better chance of the message getting through.

THE AUDIENCE BOX IS YOUR LIVE JS8CALL GROUPS

The list is read from JS8Call itself, not from a list kept inside JS8Map, so it matches whatever groups you are actually a member of. If it looks wrong or empty, JS8Map has not been able to read your JS8Call.ini — point it at the file using Browse in the control window.

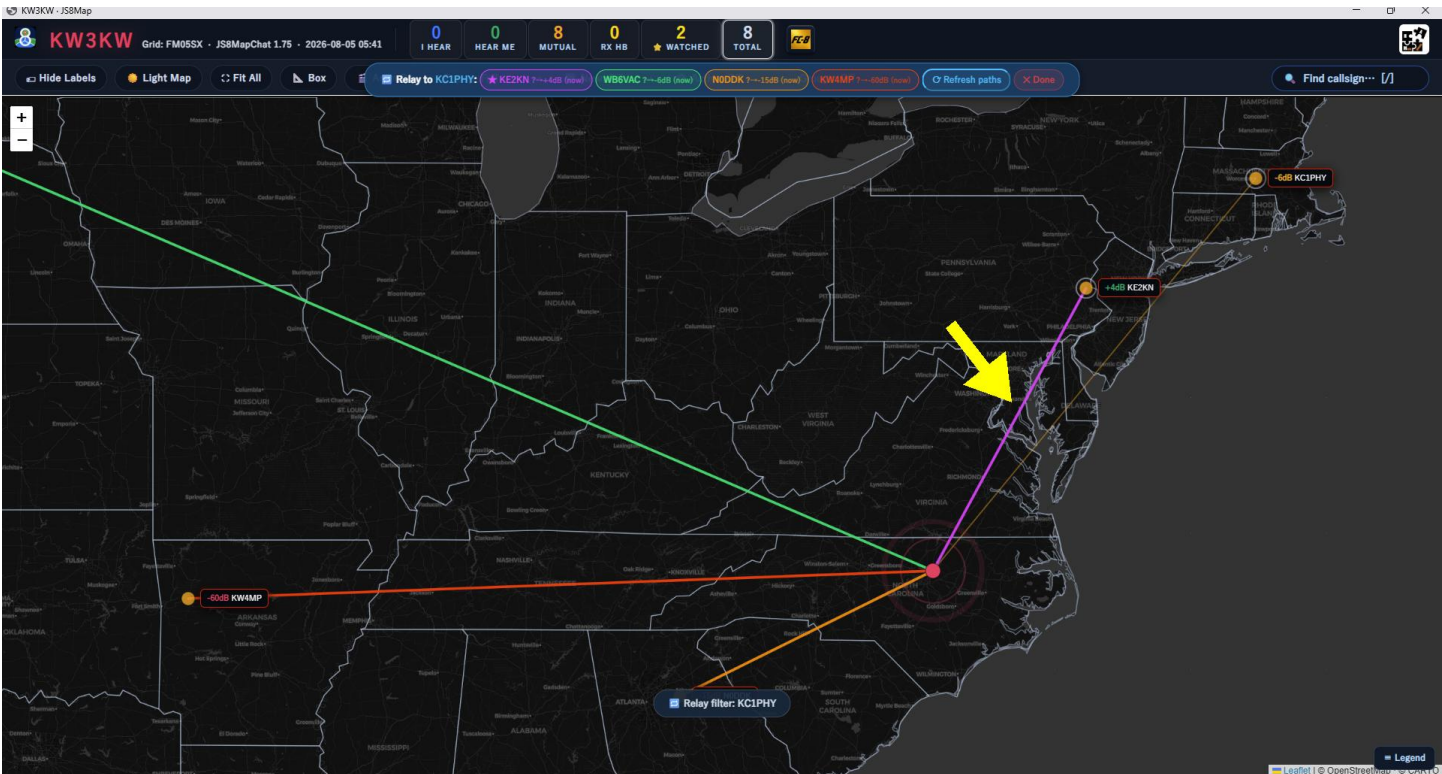


Figure 2 — The query goes out and stations begin answering.

Read the ranked answers

Stations that can reach your target reply, and within a few seconds a relay banner appears across the top of the map. Each station that answered becomes a button, ordered best first.

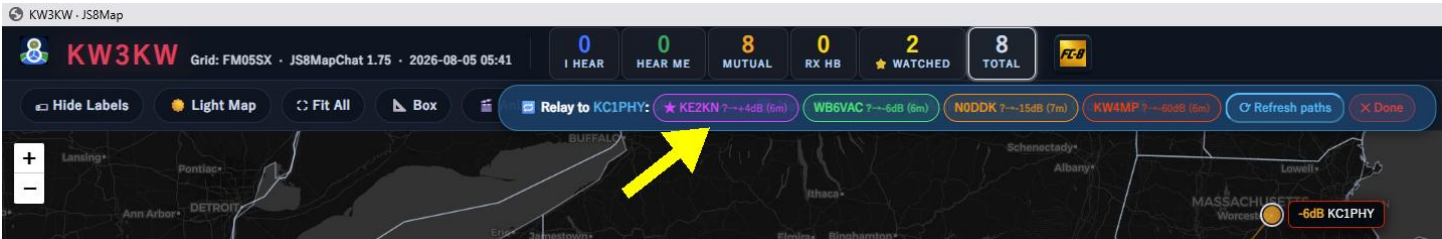


Figure 3 — The relay banner. Four stations answered for KC1PHY, ranked best first.

Read each button like this:

What you see	What it means
Star on the first button	The best path JS8Map found, by signal reports in both directions.
Button color	Rank by best outcome. Magenta is first, then green, then orange, then red.
A figure such as ?→+4dB	How the two halves of the path are reporting. A question mark means that half has not been heard yet.

What you see	What it means
A time such as (6m)	How long ago that station last confirmed it could reach the target. Fresher is better.
Refresh paths	Asks again, without starting over.
Done (finished / exit)	Leaves relay and returns the map to normal.

A PATH IS ONLY AS FRESH AS THE MINUTES BESIDE IT

These are one-off observations tied to the band as it was when the station answered. Anything much over half an hour old is treated as low confidence, and a path reported an hour ago may simply no longer exist. If the ages all look stale, click Refresh paths before committing.

Pick a station, then commit

Clicking a station draws the path it would take: a solid magenta line from you to the relay station, and a dashed magenta line onward to your target, which is shown as a diamond at its licensed position. Nothing has been sent. You can click the other stations and compare their paths freely.

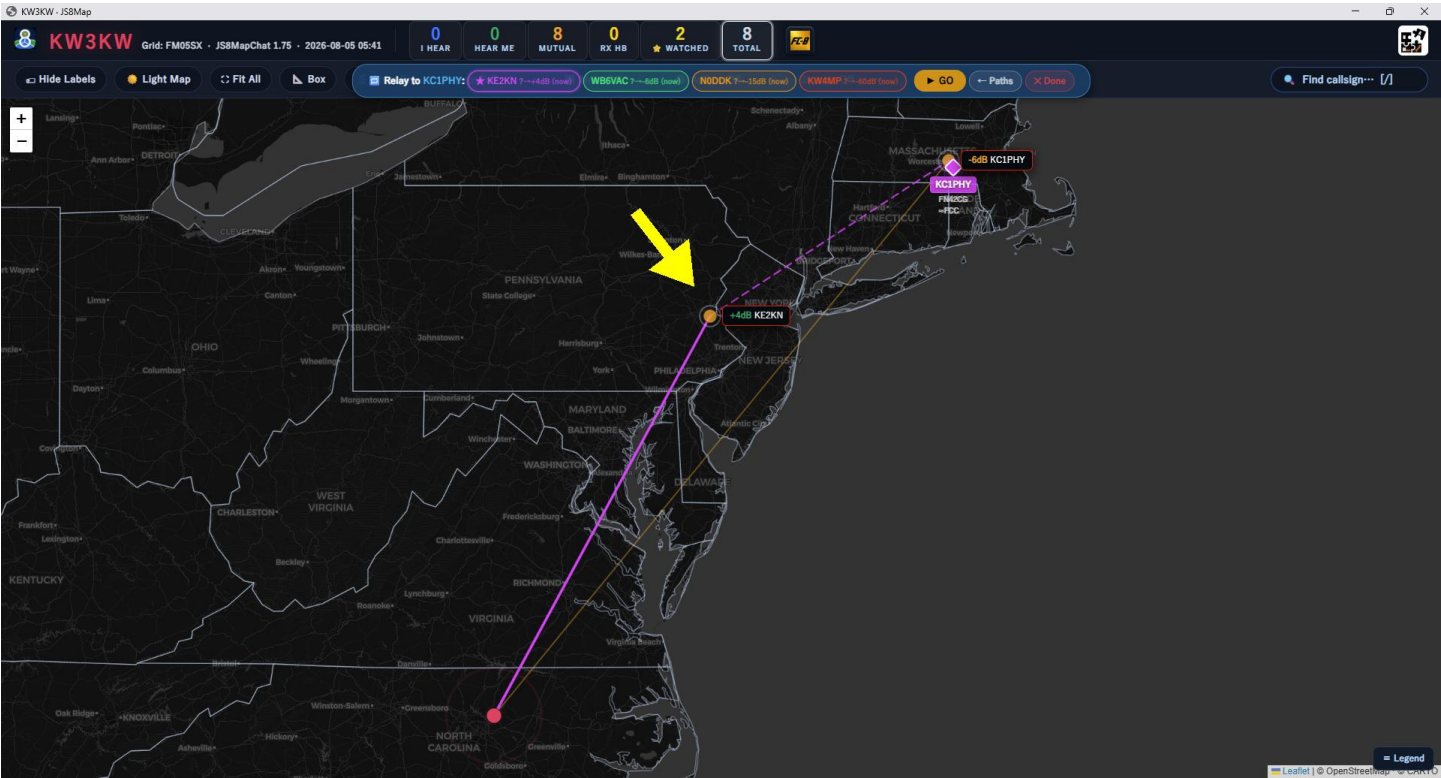


Figure 4 — The chain drawn for KE2KN. Solid magenta to the relay station, dashed onward to the target. GO has appeared.

When you are satisfied, click GO. That commits the choice, dims the stations you did not pick, and opens the message window.

SELECTING IS NOT SENDING

Clicking a station only draws its path. GO is the point of no return, and it is deliberately a second click.

Write the message and send it

The message window opens already addressed, with its title showing the route it will take and the frame it will send. Type your message and press Enter, or click Send. The frame flows through to JS8Call and goes out on the air.

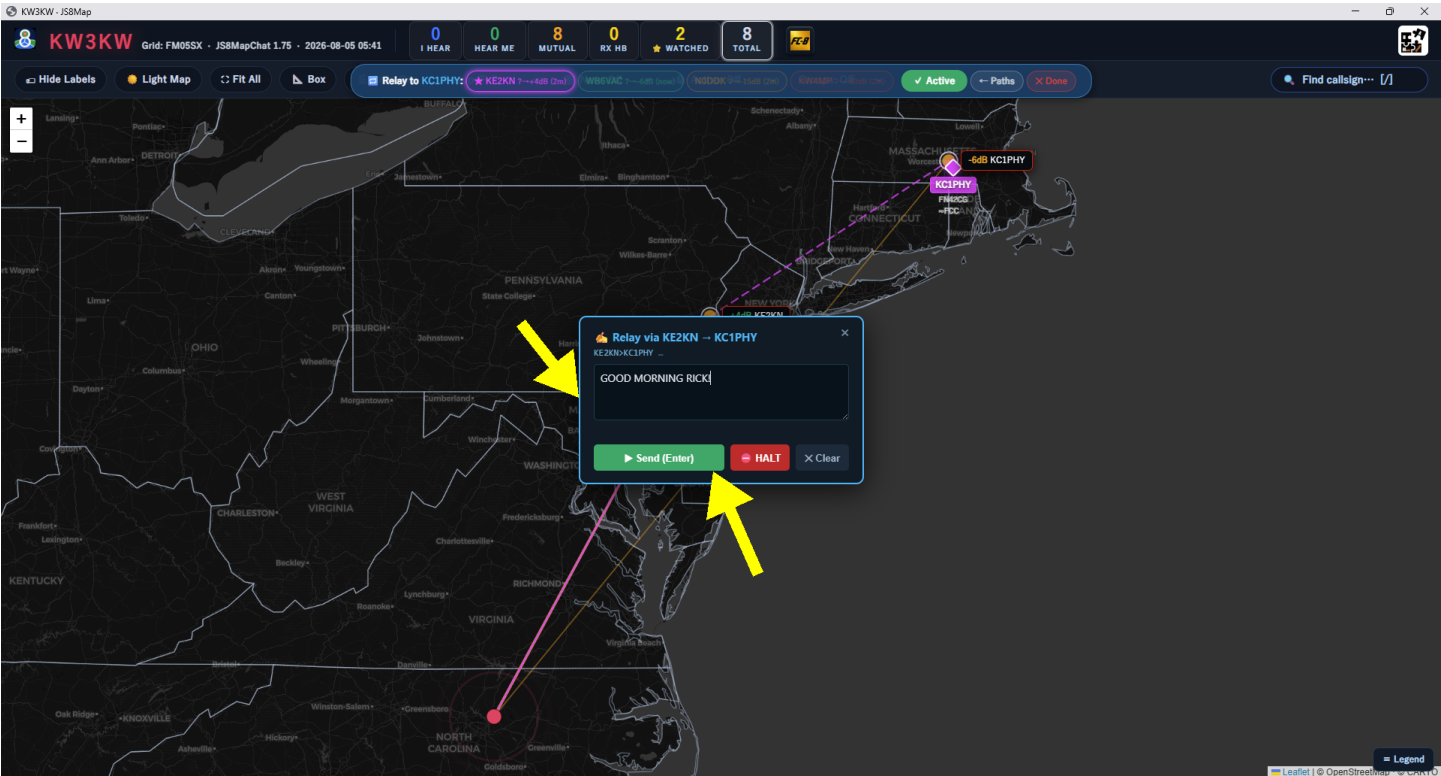
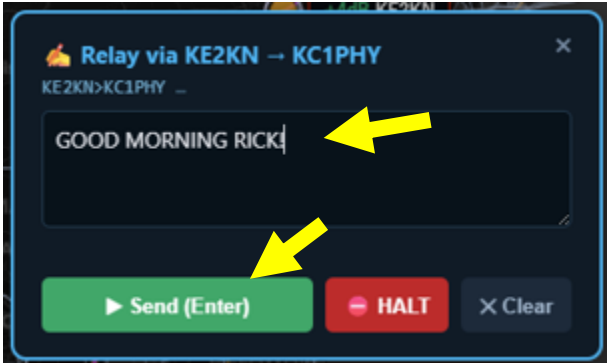


Figure 5 — The message window. The banner now reads Active and the stations you did not choose are dimmed.



Button	What it does
Send (Enter)	Sends the message with its routing attached.
HALT	Stops the transmission. The button is always shown, but it only works on JS8Call Improved 3.x. On 2.x it does nothing — use JS8Call's own Halt Tx.
Clear	Empties the box without sending.

Watching it travel

A complete relay is four transmissions, not one. Your message out to the relay station, the relay station forwarding it to your target, your target acknowledging back to the relay station, and the relay station passing that acknowledgement back to you. Each is a full JS8Call transmission, so a relay that works takes roughly three to five minutes end to end.

JS8Map draws all four. A bright dot, the comet, travels the leg that is active.

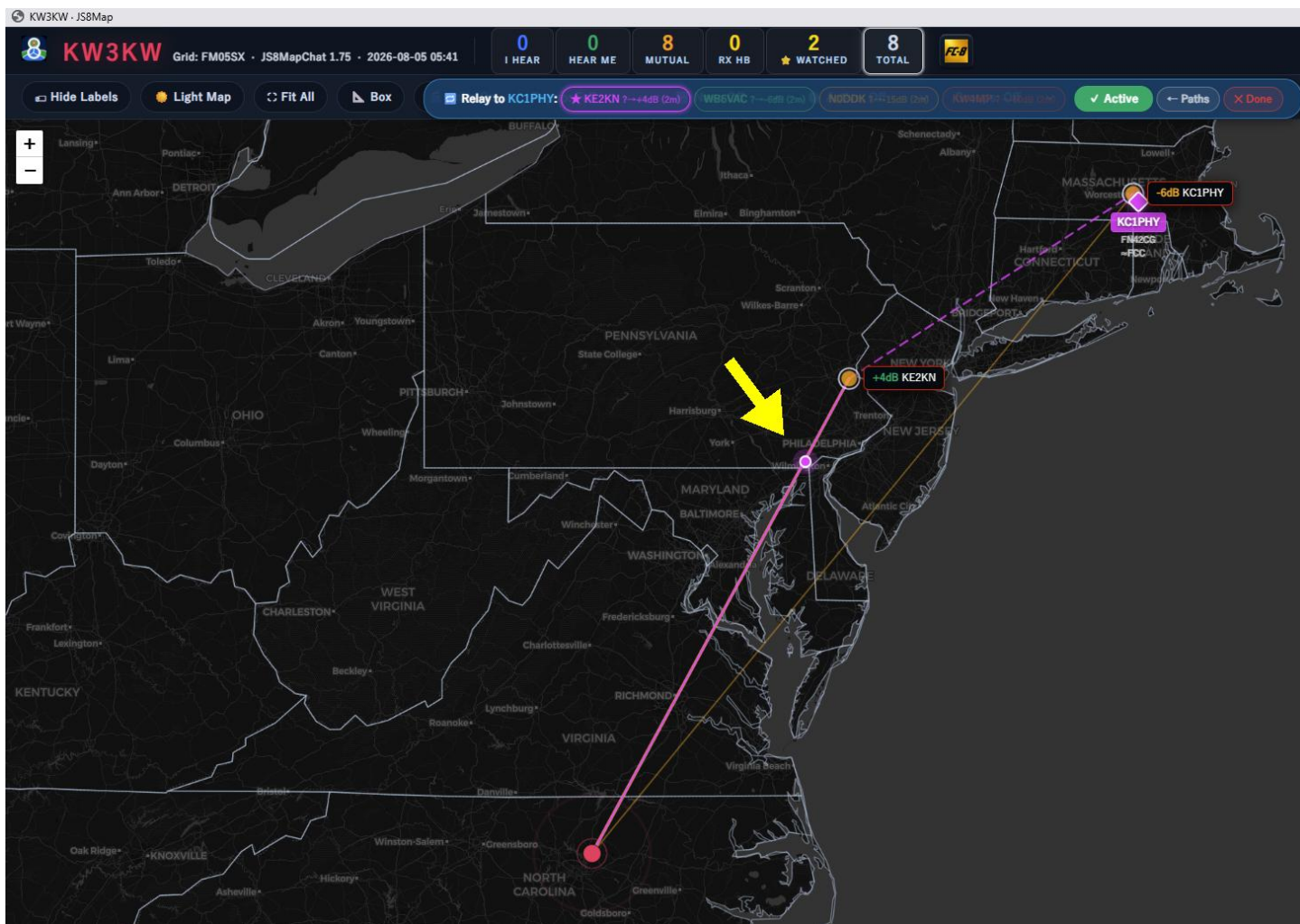


Figure 6 — Each leg is labeled with its approximate path time, out and back. The magenta comet is running the outbound leg.

What the colors mean

Two lines are drawn and each is used twice, once outbound and once on the way back. The color tells you which stage that line is at.

Color	Meaning
Magenta	Outbound. The leg your message is traveling now.
Magenta dashed	The next leg out, not started yet.
Yellow	The near leg, between you and the relay station. It turns yellow when your outbound transmission finishes, and stays yellow as the acknowledgement comes back down it.
Green dotted	The far leg, between the relay station and your target, once the acknowledgement has crossed it on the way back.
Blue comet	The acknowledgement itself, moving back toward you.

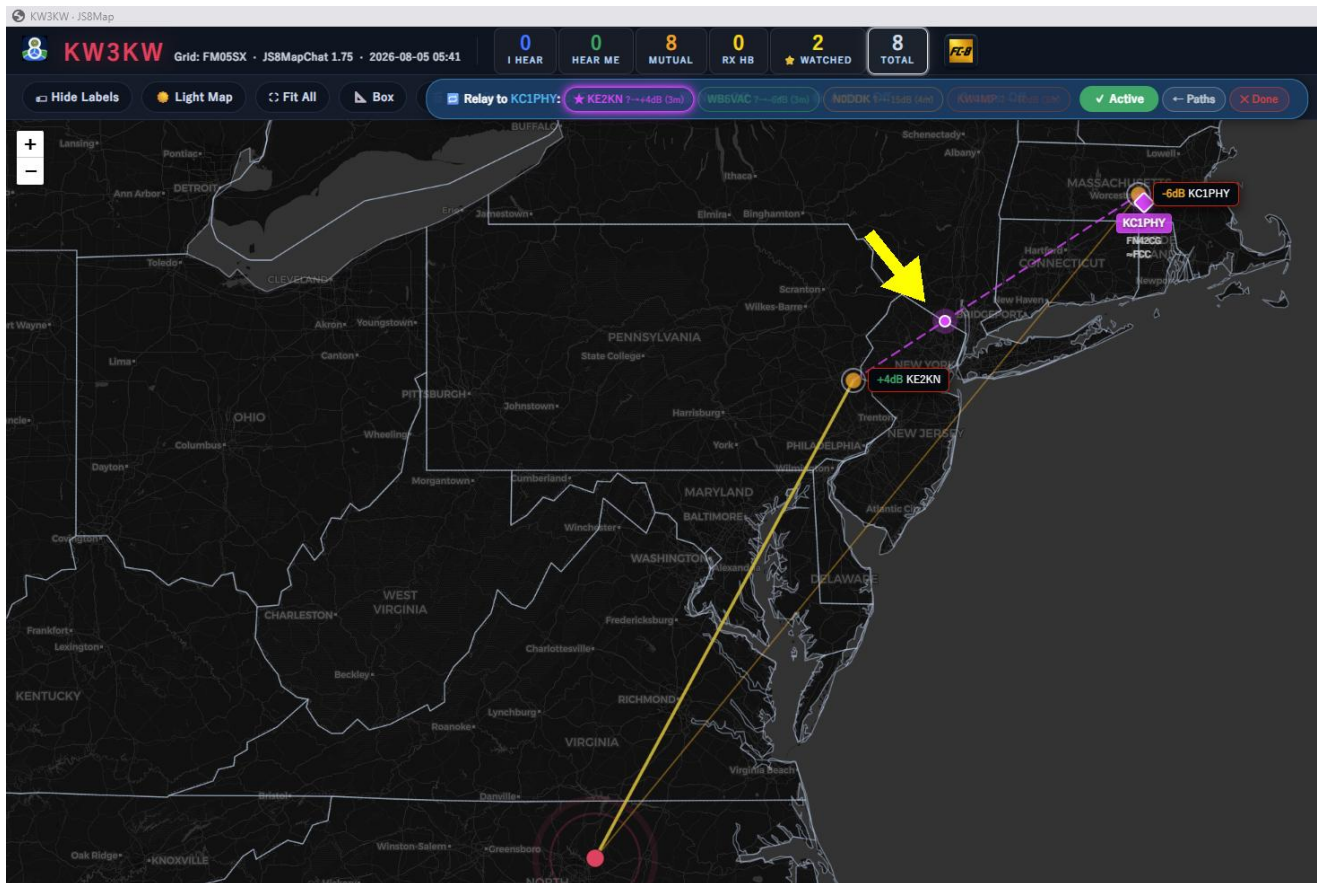


Figure 7 — A leg has finished and turned yellow. The dashed line is the leg now running.

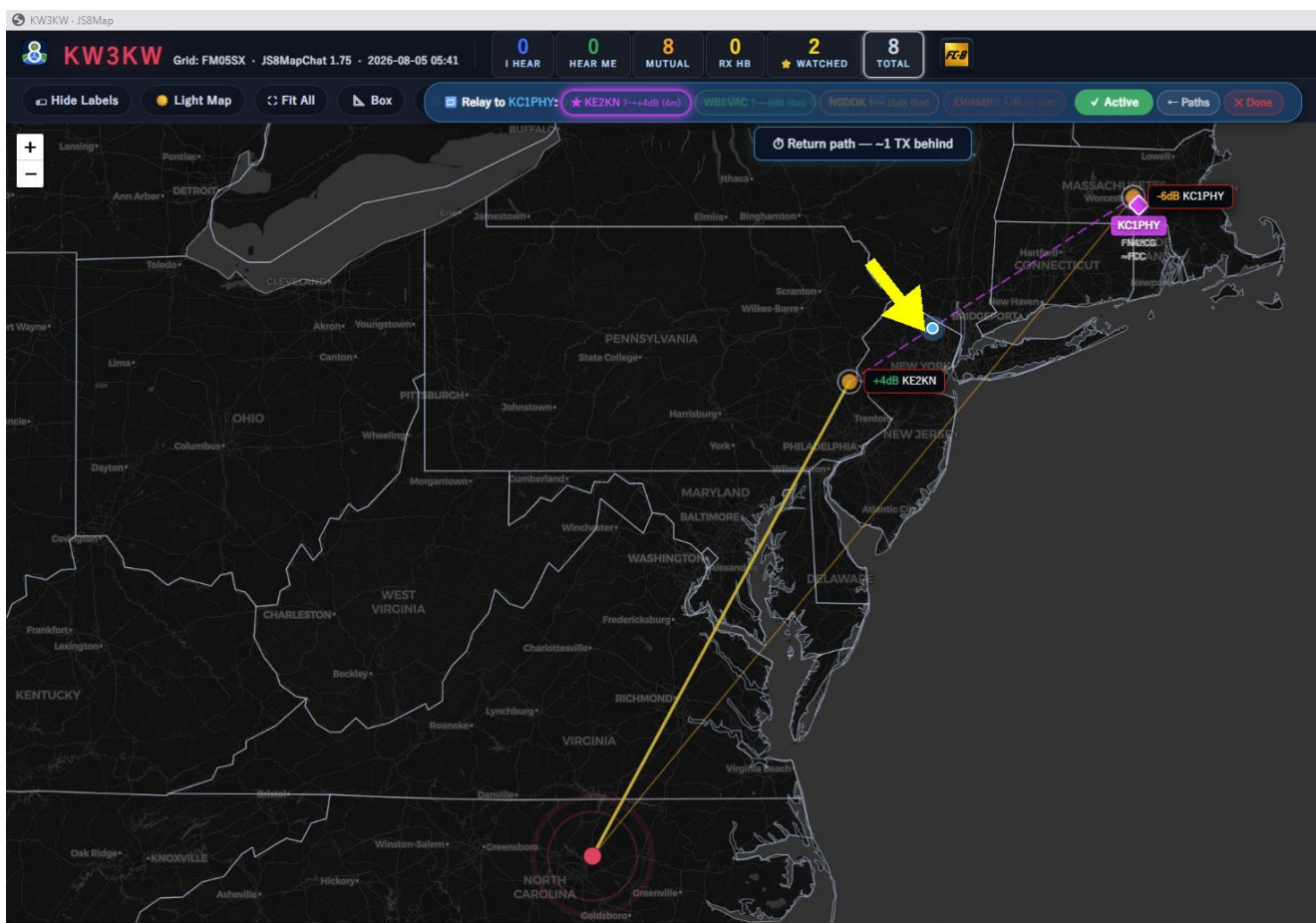


Figure 8 — The comet has turned blue: the acknowledgement is on its way back. Note the reminder that the return display runs about one transmission behind.

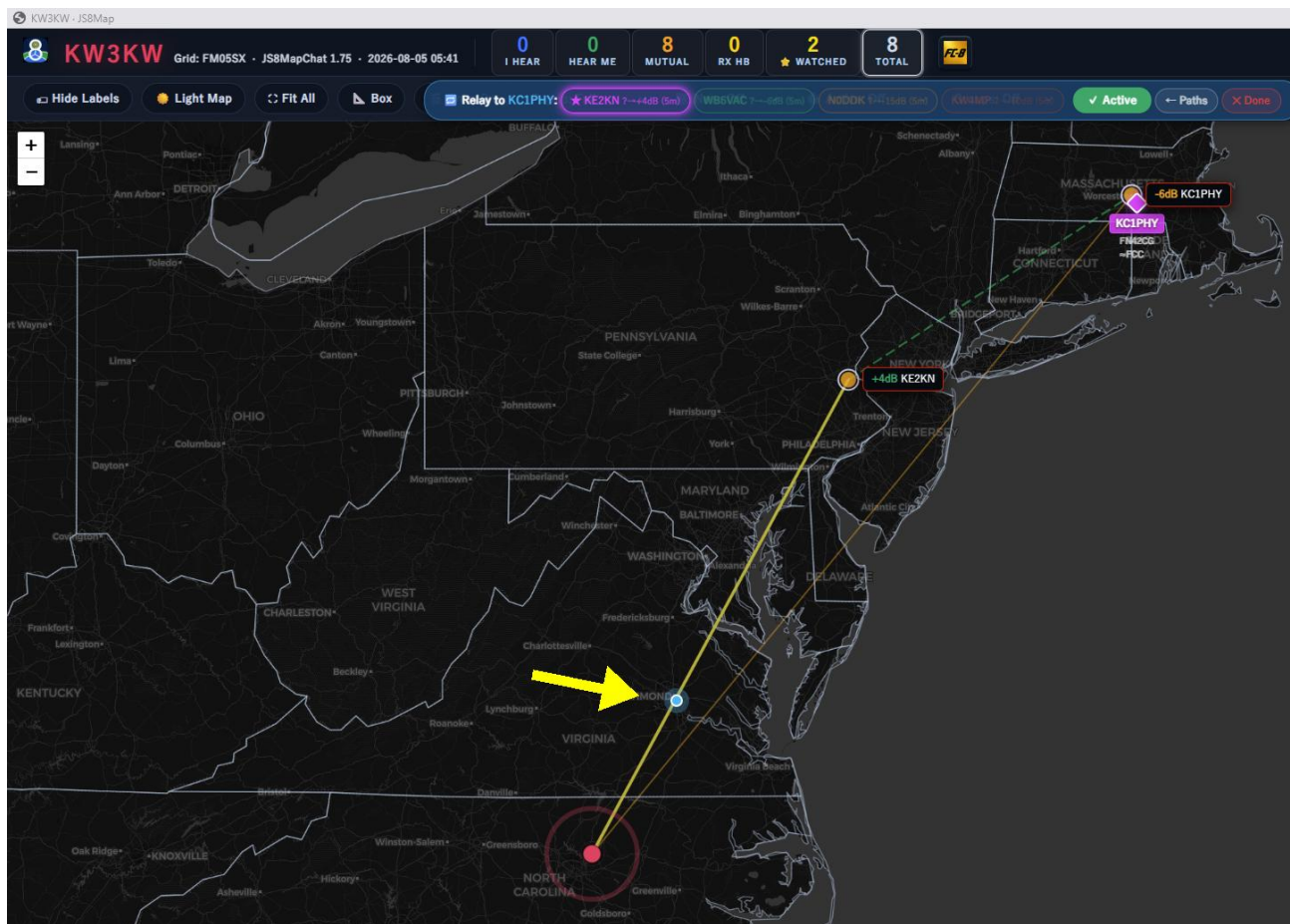


Figure 9 — The final leg, blue comet on the yellow line.

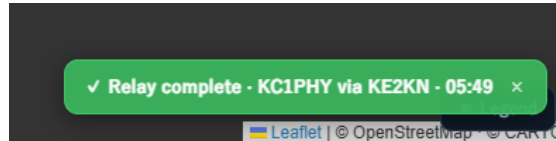


Figure 10 — Delivery confirmed.

When your target's acknowledgement reaches you, a confirmation appears at the bottom right of the map. It stays for a few minutes and then fades on its own.

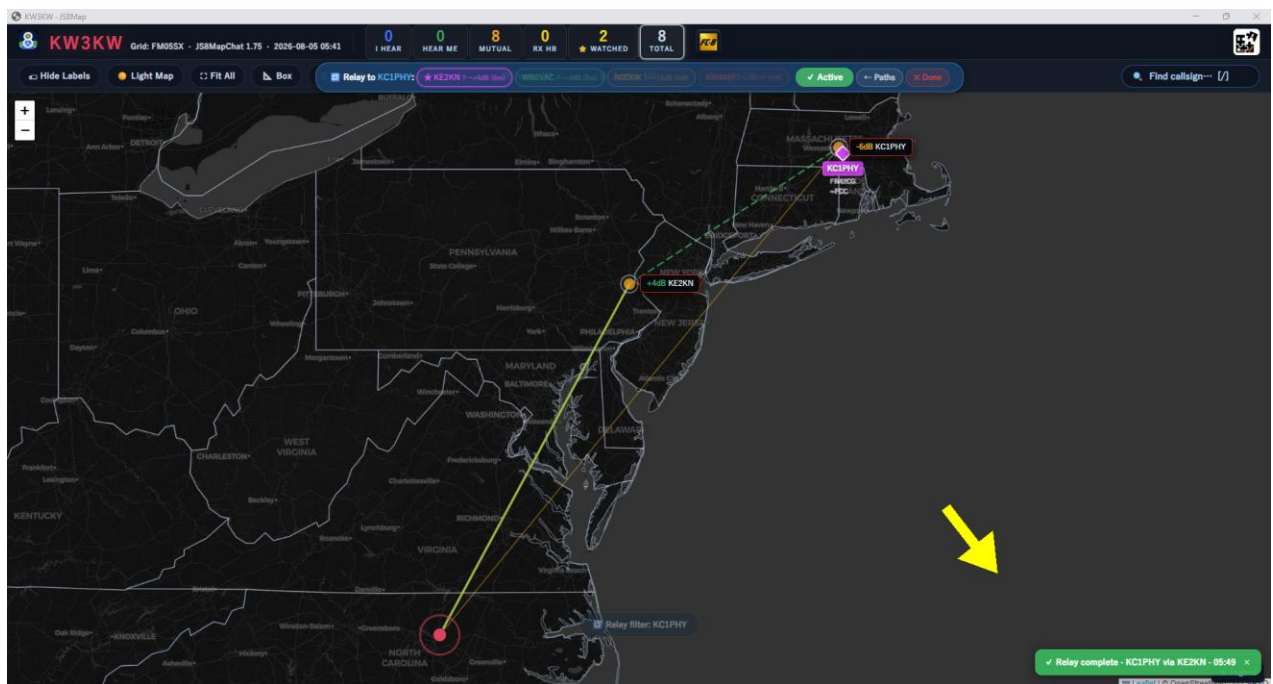


Figure 11 — Where the confirmation appears.

THE RETURN LEGS ARE ALWAYS SLIGHTLY BEHIND

JS8Map cannot know a distant station is transmitting until JS8Call has finished decoding it, which takes the whole length of that transmission. So the outbound legs are drawn as they happen, and the return legs are drawn about one transmission late. The message is further along than the screen shows. This is how JS8Call works and is not a fault of this app.

When it does not complete

If nothing comes back within about four minutes of your message being forwarded, JS8Map stops waiting and says so on the map. The legs change to amber and a warning appears at the relay station.

It means one of three things, and the display cannot tell you which:

- The relay station did not forward your message. It may not permit relay at all.
- Your target did not hear it.
- Your target heard it but does not answer automatically. Not every operator has that switched on.

Clear the warning with **Done and try a different relay station from the rankings above**. The ranked list is still there. If a late answer does arrive after the warning, the display picks up where it left off and can still finish normally.

Clear Relay ends all current paths and lets you begin a fresh search.

Relay paths are not remembered between sessions

Every time JS8Map starts, the table of relay paths is emptied.

This is deliberate. A path is a single observation tied to the band conditions at the moment it was heard, and those do not survive overnight. Keeping yesterday's answers would offer you routes that no longer exist, presented with the same confidence as fresh ones.

So the first relay of a session always begins with a Search. There is nothing to restore and nothing has been lost.